$$X_a = \frac{1}{2}v_0 t + \frac{v_0}{2\omega_1}\sin\omega_1 t - \frac{l}{2}$$
$$X_b = \frac{1}{2}v_0 t - \frac{v_0}{2\omega_1}\sin\omega_1 t + \frac{l}{2}$$
$$X_c = \frac{1}{2}v_0 t + \frac{v_0}{2\omega_2}\sin\omega_2 t + \frac{l}{2}$$
$$X_d = \frac{1}{2}v_0 t - \frac{v_0}{2\omega_2}\sin\omega_2 t + \frac{3l}{2} \qquad (4)$$

$$v_a = \frac{v_0}{2} + \frac{v_0}{2}\cos\omega_1 t$$
$$v_b = \frac{v_0}{2} - \frac{v_0}{2}\cos\omega_1 t$$
$$v_c = \frac{v_0}{2} + \frac{v_0}{2}\cos\omega_2 t$$
$$v_d = \frac{v_0}{2} - \frac{v_0}{2}\cos\omega_2 t \qquad (5)$$

1. $k_1 = k_2$

k

$$\omega_1 = \omega_2 = \sqrt{\frac{2k}{m}} = \omega_0 \qquad (6)$$

(4) a、b、c、d

(5)

$$v_a' = \frac{v_0}{2}\cos\omega_1 t$$
$$v_b' = -\frac{v_0}{2}\cos\omega_1 t$$
$$v_c' = \frac{v_0}{2}\cos\omega_2 t$$
$$v_d' = -\frac{v_0}{2}\cos\omega_2 t \qquad (2)$$

$$X_a' = -\frac{l}{2} + \frac{v_0}{2\omega_1}\sin\omega_1 t$$
$$X_b' = \frac{l}{2} - \frac{v_0}{2\omega_1}\sin\omega_1 t$$
$$X_c' = \frac{l}{2} + \frac{v_0}{2\omega_2}\sin\omega_2 t$$
$$X_d' = \frac{3l}{2} - \frac{v_0}{2\omega_2}\sin\omega_2 t \qquad (3)$$

ab M_1 X X'